

1.1: POWER FUNCTIONS

A **POWER FUNCTION** has a monomial equation (single term) of the form $y = ax^n$, where **a** is a real number and **n** is a whole number.

Ex 1: Which of the following are power functions?

a) $y = -3x^5$ b) $y = x^4 + 3x^3 - 5x^2 - 1$

c) $y = x^7$ d) $y = 2x^{-4}$

What about the function $y = 7$? Is this a power f^n ?

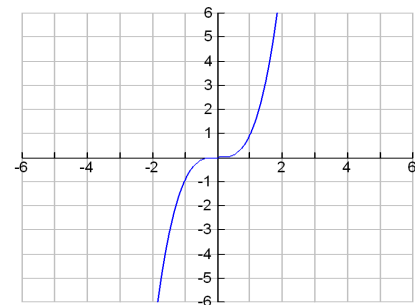
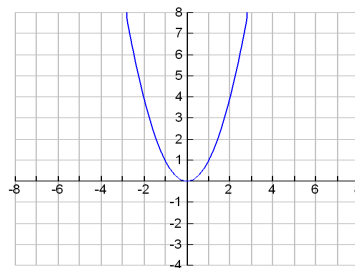
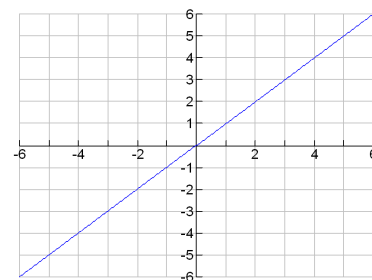
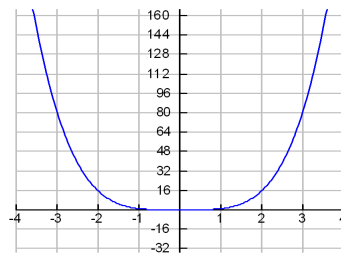
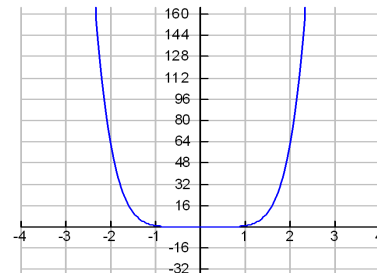
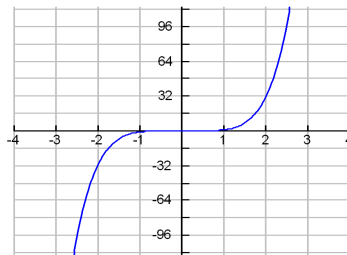
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Ex. 2:

(textbook page 5)

Given the
six power
functions,
match each
graph
with the
corresponding
equation.

- a) $y = x$
- b) $y = x^2$
- c) $y = x^3$
- d) $y = x^4$
- e) $y = x^5$
- f) $y = x^6$



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Using the graphs from the previous example, answer the following questions:

1. Which graphs have both ends extending upward in Quadrant 1 & 2?

2. What do you notice about the degree of these power functions?

3. Which graphs have one end extending downward in Q3 and the other end extending upward in Q1?

4. What do you notice about the degree of these power functions?

Summary

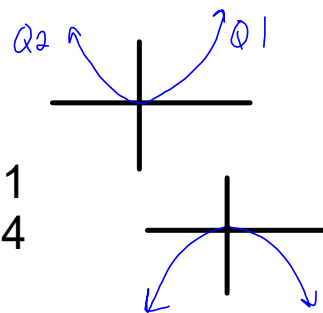
Given a power function $y = ax^n$:

Even-degree functions

if $a > 0$, the graph extends from Q2 to Q1

if $a < 0$, the graph extends from Q3 to Q4

line symmetry in the y-axis

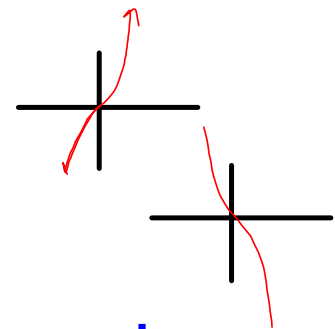


Odd-degree functions

if $a > 0$, the graph extends from Q3 to Q1

if $a < 0$, the graph extends from Q2 to Q4

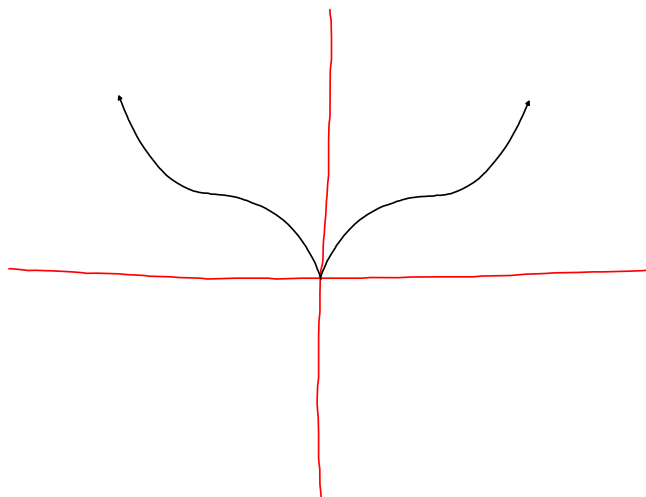
point symmetry about the origin



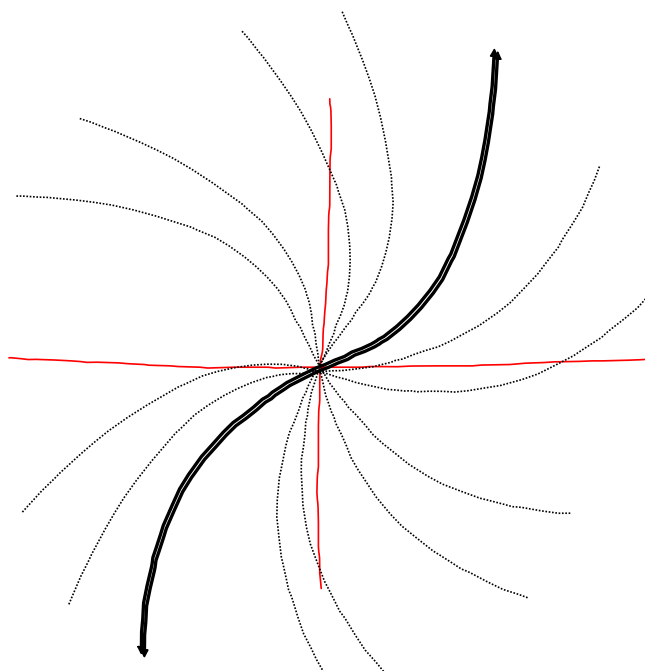
Even-degree power functions have line symmetry in the line $x = 0$, (reflection in the y-axis).

Odd-degree power functions have point symmetry about the origin, (can be rotated 180° from the point $(0,0)$).

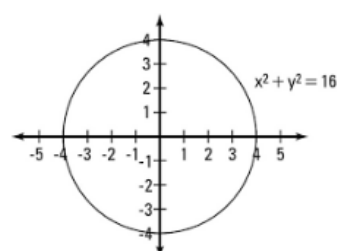
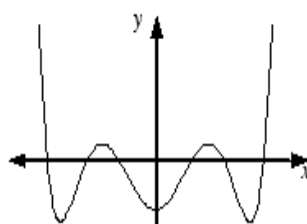
Line symmetry: reflection in y-axis



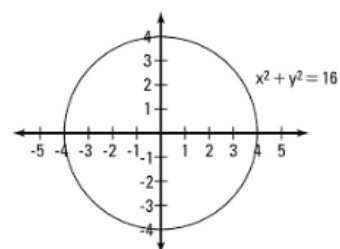
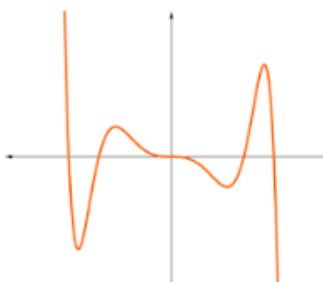
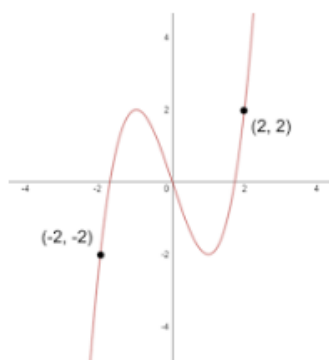
Point symmetry: 180° rotation counter-clockwise



Line symmetry: reflection in y-axis

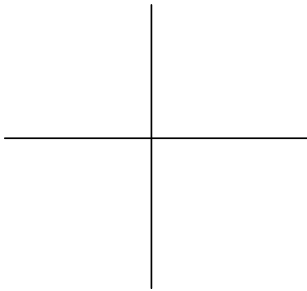


Point symmetry: 180° rotation counter-clockwise

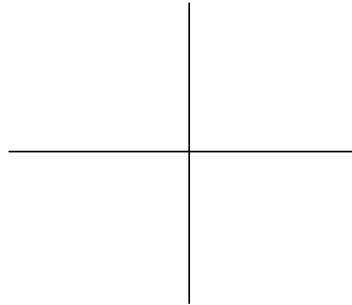


Ex 3: Draw a sketch of the following power functions.
Describe the **end behaviour** and state the domain and range for each.

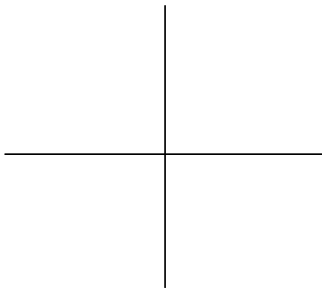
a) $y = 3x$



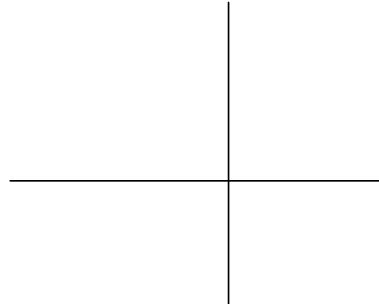
b) $y = 2x^4$



c) $y = -x^8$



d) $y = -7x^5$



HMWK: p.12 # 3-5, 7,10-12,14

3d (ii), (iii) and (v) answer is incorrect